

BÁLINT LÁSZLÓ SZARVAS

From: Hungary ◇ Since: 11/20/1999 ◇ szarvas.babi@gmail.com ◇ github.com/balintszarvas

EDUCATION AND TRAINING

PhD in Artificial Intelligence

July 2026 – Current

ELLIS Unit Linz, JKU

- Starting my ELLIS PhD in AI for biological applications under Prof. Günter Klambauer, Prof. Sepp Hochreiter and one more supervisor TBD at another institutiton

MSc Thesis Exchange

Mar 2025 – Mar 2026

ETH Zürich

- Research at the Distributed Computing Lab under Prof. Roger Wattenhofer, Florian Grötschla, Joël Mathys and Dr. Valeria Krzhizhanovskaya (UvA). Title: *Cross-Scale Interaction of Components in Neural Closure Models for Weather*
- Inspired by the recent success of hybrid (Physics Solver + ML) models in weather prediction, I developed a model that, in addition to competitive predictive skill, enables the empirical study of the interaction between its components.

MSc Computational Science

Sep 2023 – June 2026

University of Amsterdam / Vrije Universiteit Amsterdam (joint degree)

- Courses taken: Stochastic Simulation, Machine Learning for Health Data, Distributed Systems, Numerical Algorithms, Data Mining Techniques, Scientific Computing, Theory of Complex Systems, Complex Systems Simulation.

BSc Business Administration and Management

Aug 2018 – Jan 2022

Corvinus University of Budapest

- Majored in Entrepreneurship. Courses in Probability Theory, Linear Algebra, Statistics, Economics, Finance, Management, Process Optimisation, Operations Research.
- Thesis title: *Circular Business Models for Startups* (grade 5.00/5.00), Final GPA: 4.56/5.00

WORK EXPERIENCE

Student Researcher

Nov 2024 – May 2025

Massivizing Computer Systems Group, Vrije Universiteit Amsterdam

- Research alongside my university coursework under Prof. Alexandru Iosup. We developed *FAILS*: A Framework for Automated Collection and Analysis of LLM Service Incidents, providing statistical insights into the reliability of popular LLM services.

Student Researcher

Aug 2024 – Feb 2025

Amsterdam Machine Learning Lab, University of Amsterdam

- Research alongside my university coursework under Maksim Zhdanov. We propose Conditional Clifford-Steerable CNNs, conditioning the convolution operator on auxiliary signals to yield an expressive equivariant CNN family that achieves state-of-the-art results on diverse PDE-learning benchmarks.

Research Assistant – Computational Neuroscience

Jan 2023 – May 2023

August Pi i Sunyer Biomedical Research Institute, Barcelona, Spain

- Under Prof. Pablo Jercog researching spatial information encoding in the hippocampus. Analised data and tracked trajectories of mice using deep learning, contributing to a publication ([Link](#)) (not listed as author by prior agreement due to the indirect nature of my contribution).

Product Intern

Mar 2022 – Dec 2022

Braingaze, Barcelona, Spain

- Internship at a neurotechnology startup led by Prof. Hans Supér. Responsibilities included utilising classification algorithms to detect Alzheimer's disease and ADHD based on saccadic eye-movements, and ensuring our software met the needs of both physicians and patients.

PUBLICATIONS

Conditional Clifford-Steerable CNNs for PDE Modeling

Authors: **Bálint László Szarvas***, Maksim Zhdanov*

Venue: ICML 2026, AI4Science Workshop @ NeurIPS 2025 preprint: arXiv:2510.14007

FAILS: A Framework for Automated Collection and Analysis of LLM Service Incidents

Authors: **Bálint László Szarvas***, Sándor Battaglini-Fischer*, Nishanthi Srinivasan*, Xiaoyu Chu, Alexandru Iosup

Venue: 8th Workshop on Hot Topics in Cloud Computing Performance @ ICPE 2025 link: ACM Digital Library

HONOURS AND AWARDS

STIPI Talent Recognition

Hungarian National Talent Center

Jul 15, 2023

- Given to the nation's 50 best students pursuing studies at prestigious universities abroad. Link

2nd Place at Hacking the Future of HealthTech

Xartec Salud, Polytechnic University of Catalonia

Nov 6, 2022

- Developed a cost-effective vision-aiding VR tool for visually impaired patients. Link

National Scholarship for BSc

Government of Hungary

Sep 1, 2018

- Full, merit-based scholarship covering tuition fees of my BSc degree.

SKILLS

Languages: Hungarian (Native), English (C2), Spanish (B2)

Programming: Python, MATLAB

Data Science and ML: Numpy, JAX, PyTorch

EXTRACURRICULAR ACTIVITIES

Neuro-AI Summer School

Amsterdam Brain and Cognition Center

Jul 2024

- Taught by researchers from Oxford University, UCL and Imperial College on reinforcement learning, neural decoding and cognitive maps.

Talk that Science

Sep 2024 – Feb 2025

- Podcast Host and Science Communicator. Link

Middle Manager at AIESEC Hungary

Sep 2019 – Jul 2021

- Middle manager at the largest student-led organisation in the world, providing exchange opportunities to students of all countries. Link

D1 Sprint Kayaker

Jul 2014 – Jul 2015

- 2nd and 3rd place at the National Junior Championships in K2 and K1 relay.

Musician

For life

- I write, produce songs and play the drums.

References available upon request